Product and Quotient rule

In this session[30] we are going to learn about the product and quotient rule for derivatives. This rule allow us to compute the derivative of a product of functions. Assuming that we have a function $f(x) = u(x)v(x)$, the derivative of the function can be computed as follows,

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left[u(x)v(x)\right] = \frac{\mathrm{d}u(x)}{\mathrm{d}x}v(x) + u(x)\frac{\mathrm{d}v(x)}{\mathrm{d}x}. \tag{3}$$

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For example, if we need to compute the derivative of $f(x) = x^2 \sin(x)$, we can compute it as follows,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left[x^2\sin(x)\right] &= \left[\frac{\mathrm{d}}{\mathrm{d}x}x^2\right]\sin(x) + x^2\left[\frac{\mathrm{d}}{\mathrm{d}x}\sin(x)\right] \\ &= [2x]\sin(x) + x^2\left[\cos(x)\right] \\ &= 2x\sin(x) + x^2\cos(x).\end{aligned}$$

On the other hand, the quotient rule allow us to easily compute the derivative of a quotient of functions. In this instance the function is of the form $f(x) = \dfrac{u(x)}{v(x)}$. In order to compute it's derivative we use the following rule,

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \frac{\mathrm{d}}{\mathrm{d}x}\left[\frac{u(x)}{v(x)}\right] = \frac{1}{v^2(x)}\left[\frac{\mathrm{d}u(x)}{\mathrm{d}x}v(x) - u(x)\frac{\mathrm{d}v(x)}{\mathrm{d}x}\right]. \tag{4}$$

For example, if we are asked to calculate the derivative of $f(x) = \dfrac{x^2}{\sin(x)}$ we can do it as follows,

$$\begin{aligned}\frac{\mathrm{d}f(x)}{\mathrm{d}x} &= \frac{1}{\sin^2(x)}\left[\left(\frac{\mathrm{d}}{\mathrm{d}x}x^2\right)\sin(x) - x^2\left(\frac{\mathrm{d}}{\mathrm{d}x}\sin(x)\right)\right] \\ &= \frac{1}{\sin^2(x)}\left[(2x)\sin(x) - x^2\left(\cos(x)\right)\right] \\ &= \frac{2x}{\sin(x)} - \frac{x^2}{\sin(x)}\cot(x) = x\csc(x)\left[2 - x\cot(x)\right]\end{aligned}$$

Class activity

Compute the derivatives uing the product, quotient and power rules of the following functions. You can also use the rules for trascendental functions if necessary.

$$\begin{aligned}f(x) &= x^3e^x \\ g(x) &= \frac{\sin(x)}{x} \\ h(x) &= x^2\ln(3x)\end{aligned}$$

Derivation of the Product rule

In today's session[31] we are going to learn where the product and quotient rules for derivatives come from.

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Starting with the product rule, we recall that $f(x) = u(x)v(x)$. Then, by using the definition of the derivative, we get

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h\to 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h}.$$

In order to get somewhere, let's add a clever 0, $u(x+h)v(x) - u(x+h)v(x)$,

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h\to 0} \frac{u(x+h)v(x+h) - u(x+h)v(x) - u(x)v(x) + u(x+h)v(x)}{h}.$$

This clever 0 allow us to factor terms,

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h\to 0} \left[u(x+h)\frac{v(x+h) - v(x)}{h} + v(x)\frac{u(x+h) - u(x)}{h} \right].$$

By applying the limit and using properties,

$$\begin{aligned} \lim_{h\to 0} \left[u(x+h)\frac{v(x+h) - v(x)}{h} \right] &= \left[\lim_{h\to 0} u(x+h) \right] \left[\lim_{h\to 0} \frac{v(x+h) - v(x)}{h} \right] \\ &= u(x)\frac{\mathrm{d}v(x)}{\mathrm{d}x} \end{aligned}$$

$$\begin{aligned} \lim_{h\to 0} \left[v(x)\frac{u(x+h) - u(x)}{h} \right] &= \left[\lim_{h\to 0} v(x+h) \right] \left[\lim_{h\to 0} \frac{u(x+h) - u(x)}{h} \right] \\ &= v(x)\frac{\mathrm{d}u(x)}{\mathrm{d}x} \end{aligned}$$

We show that,

$$\frac{\mathrm{d}f(x)}{\mathrm{d}x} = \lim_{h\to 0} \frac{u(x+h)v(x+h) - u(x)v(x)}{h} = u(x)\frac{\mathrm{d}v(x)}{\mathrm{d}x} + v(x)\frac{\mathrm{d}u(x)}{\mathrm{d}x}.$$

Class activites

Compute the derivatives using the product, quotient and power rules of the following functions. You might need the rules of transendental functions.

$$\begin{aligned} f(x) &= \frac{\ln(x)}{e^x} \\ g(x) &= e^x \ln[x] \sin(x) \\ h(x) &= \ln(x)\sin(x) \end{aligned}$$